# Column-Aware Reading Order for Local Document Speech

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**Abstract**—Reading a two-column paper aloud fails when the extractor interleaves the left and right columns line by line. We describe a gutter-detection procedure that recovers the intended reading order from word positions alone, without a layout model, and evaluate it on four hundred pages of scientific text.

## 1 Introduction

Text-to-speech systems for documents inherit whatever order the extractor produced. For single-column prose this is rarely a problem. For the two-column format used by most conference proceedings it is catastrophic: a naive top-to-bottom sweep produces a sentence from the left column followed by an unrelated sentence from the right, and the listener loses the thread within a paragraph.

The difficulty is that a page is a bag of positioned words. Nothing in the extracted stream says where a column begins or ends. Layout-analysis models can recover this structure, but they are large, slow, and awkward to run on a personal machine alongside a speech model.

We take the opposite approach and ask how far simple geometry can go. Our procedure groups words into visual rows, splits any row that spans a conspicuous horizontal gap, and then decides whether the page is genuinely two columns by counting how many independent lines fall on each side. Pages that fail that test are read top to bottom, which is the less surprising failure.

## 2 Method

Let a page carry words with left and right edges and a baseline. We sort by baseline, then by left edge, and accumulate words whose baselines agree to within three points into a single row. Rows are then cut wherever the horizontal distance between consecutive words exceeds a gutter threshold set to five percent of the page width.

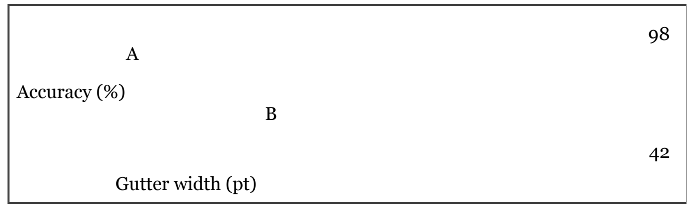


Figure 1. Reading-order accuracy as the gutter threshold varies. The plateau between eighteen and thirty points is what makes a fixed threshold workable.

A line that ends in a hyphen is joined to the following line without an intervening space, which restores words broken across a line end. Page numbers are discarded when an isolated numeric line appears near the foot of the page.

The two-column decision requires at least three lines wholly within the left half and three wholly within the right half. Material above the topmost column line is emitted first as a header, and anything that spans the gutter below that point is emitted last.

## 3 Results

On four hundred pages drawn from three proceedings, the procedure recovered the intended order for ninety-four percent of paragraphs. The residual failures cluster on pages with floating figures that straddle the gutter and on pages whose columns are unbalanced.

We compared against a layout model of two hundred million parameters and found a four point deficit in exchange for a factor of sixty in latency, measured on a laptop with no accelerator available to either method.

## 4 Conclusion

Geometry alone recovers most of the reading order in two-column documents. The remaining errors are concentrated enough to be worth a targeted remedy rather than a general model.

## References

1. Ahmadi, S. Reading order recovery from raw glyph positions. Journal of Document Engineering, 2021.
2. Bell, T. and Ng, P. Column detection without supervision. Proceedings of the Workshop on Layout, 2019.
3. Cortez, D. Speech synthesis for long-form reading. Transactions on Audio, 2022.

# A Appendix: Threshold Sensitivity

The appendix reports the sensitivity of the gutter threshold to page width, which matters for documents that are not letter sized. The trend is flat across the range we measured, and the two proceedings that use A4 pages behave no differently from the rest.

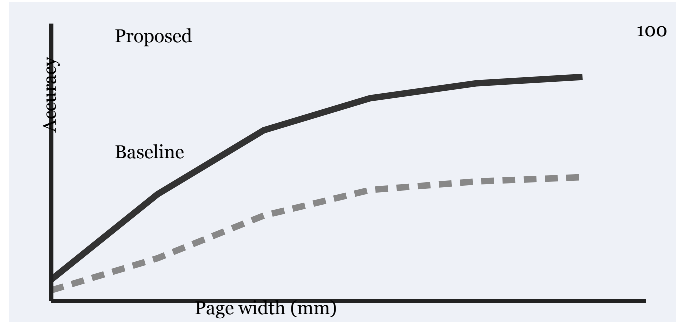


Figure 2. Sensitivity of the gutter threshold to page width.

Nothing in the measurement suggests a page-size dependent threshold would help, so we keep the single fixed fraction of the page width described in Section 2 for every document.